

TARK Vision App

TARK VISION

NO SIGNAL

▶ Start

■ Stop

Rotat
e

📶 Connect

📶 Disconnect

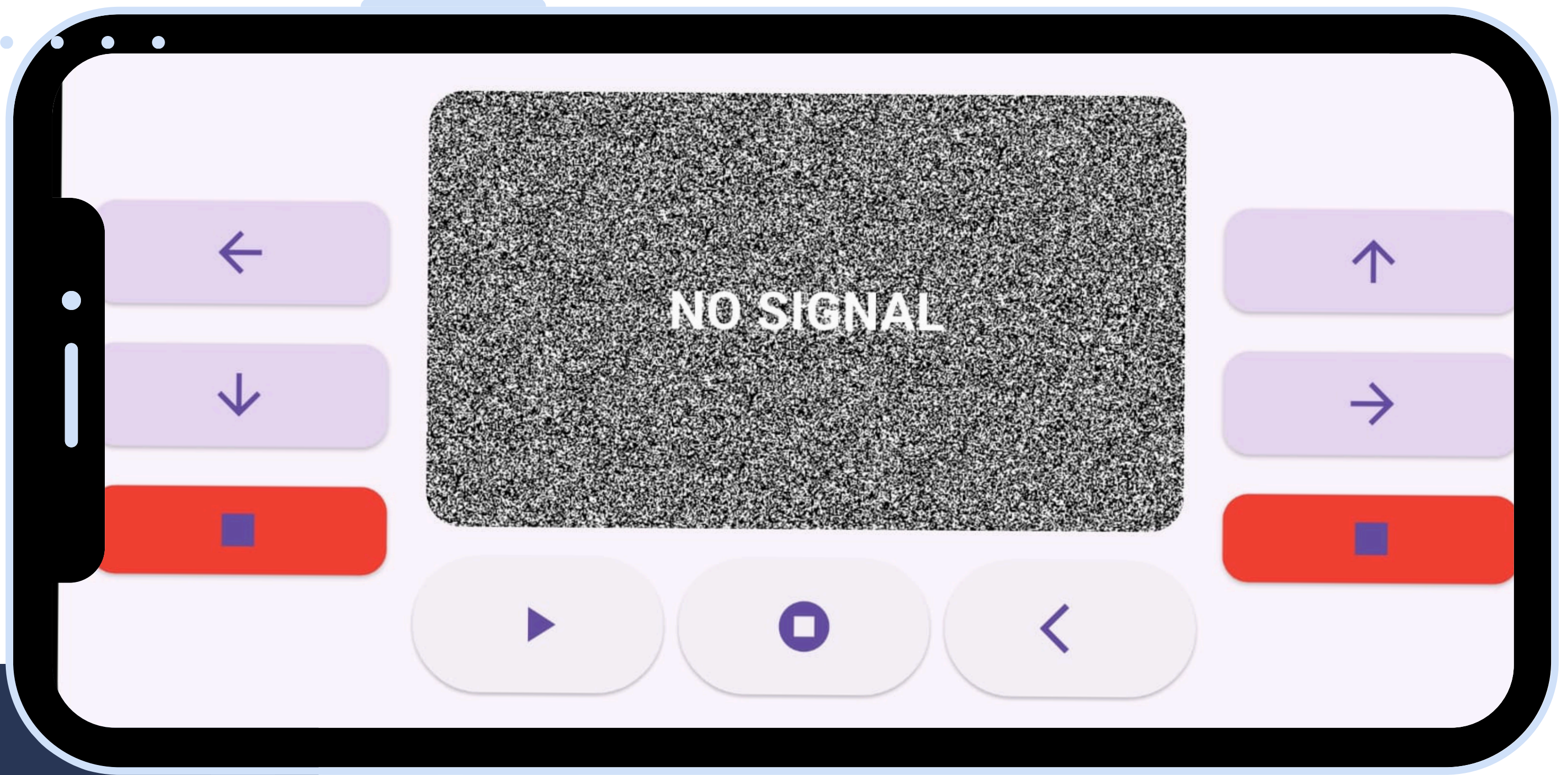
Disconnected

APP FEATURES:

- DUAL LAYOUT: PORTRAIT MODE AND LANDSCAPE MODE.
- PORTRAIT MODE HAS LIVE STREAM CONTAINER, START/STOP STREAM ICON BUTTONS, CONNECT/DISCONNECT. SWITCH TO LANDSCAPE MODE.
- LANDSCAPE MODE: LIVE STREAM CONTAINER, START/STOP STREAM ICON BUTTONS, ROBOTIC CAR CONTROLS(FORWARD, BACKWARD, LEFT, RIGHT AND STOP).
- A FALLBACK WHITENOISE DISPLAY.

STEPS FOR CONNECTION:

- POWER ON ESP32 CAMERA MODULE.
- CLICK RESET
- GO TO MOBILE WIFI SETTINGS-> CONNECT TO "VISION" ENTER PASSWORD: 12345678
- OPEN TARKVISION APP: CLICK ON CONNECT. A MESSAGE WILL BE DISPLAYED ON SUCCESSFUL CONNECTION
- CLICK ON START STREAM ICON BUTTON LIVE STREAM WILL BE VISIBLE IN THE CONTAINER.
- CLICK ON ROTATE-> SWITCHES TO LANDSCAPE MODE.
- TRY ROBOTIC CAR CONTROLS: UP, DOWN, LEFT, RIGHT AND STOP ICON BUTTONS.



ARCHITECTURE:

- ESP32 CAMERA MODULE (ACCESS POINT MODE)
- HTTP CONNECTION PROTOCOL(REQUEST / RESPONSE)
- USES DUAL PORTS 80(DEFAULT) AND 81 FOR LIVE STREAMING AND ROBOTIC CONTROLS RESPECTIVELY.
- APP (CONNECTED TO ESP32 CAMERA) (STATION MODE)
- MOTOR DRIVER FOR CONNECTING MOTORS WITH ESP32 CAMERA MODULE.

ADVANTAGES:

- DUAL MODE WITH LIVE STREAM CONTAINER.
- COMPACT AND LIMITED HARDWARE.
- USES DUAL PORT FOR ROBOTIC CONTROLS AND LIVE STREAMING THUS SMOOTHER ACCESS TO ESP32 CAMERA MODULE SERVER.

LIMITATIONS:

- TWO WHEELED DRIVE DUE TO POWER AND PIN CONSTRAINTS.
- LIMITED RANGE. CAN BE INCREASED USING WIFI MODULE ANTENNAS.